Лабораторная работа №2

Настройка DNS-сервера

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Информация

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Цели и задачи

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- Приобретение практических навыков по установке и конфигурированию DNS-сервера, усвоение принципов работы системы доменных имён.

1. Установите на виртуальной машине server DNS-сервер bind и bind-utils (см. раздел 2.4.1).
2. Сконфигурируйте на виртуальной машине server кэширующий DNS-сервер (см. раздел 2.4.2).
3. Сконфигурируйте на виртуальной машине server первичный DNS-сервер (см. раздел 2.4.3).
4. При помощи утилит dig и host проанализируйте работу DNS-сервера (см. раздел 2.4.4).
5. Напишите скрипт для Vagrant, фиксирующий действия по установке и конфигурированию DNS-сервера во внутреннем окружении виртуальной машины server. Соответствующим образом внесите изменения в Vagrantfile (см. раздел 2.4.5).

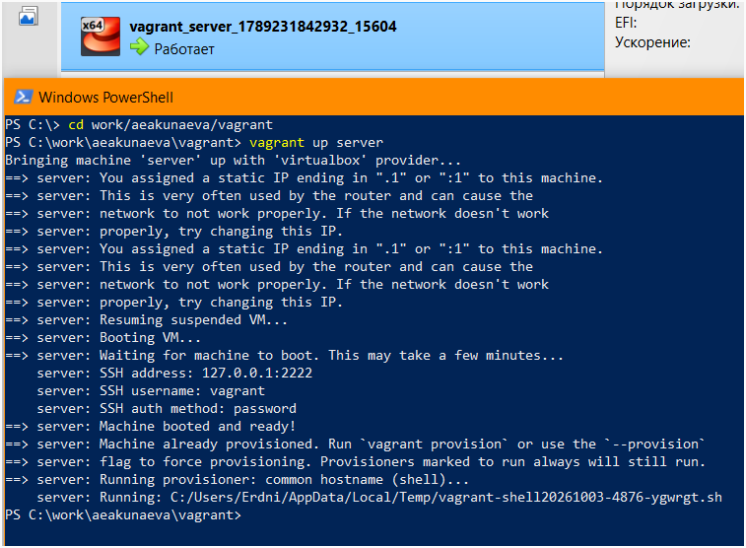
Материалы и методы

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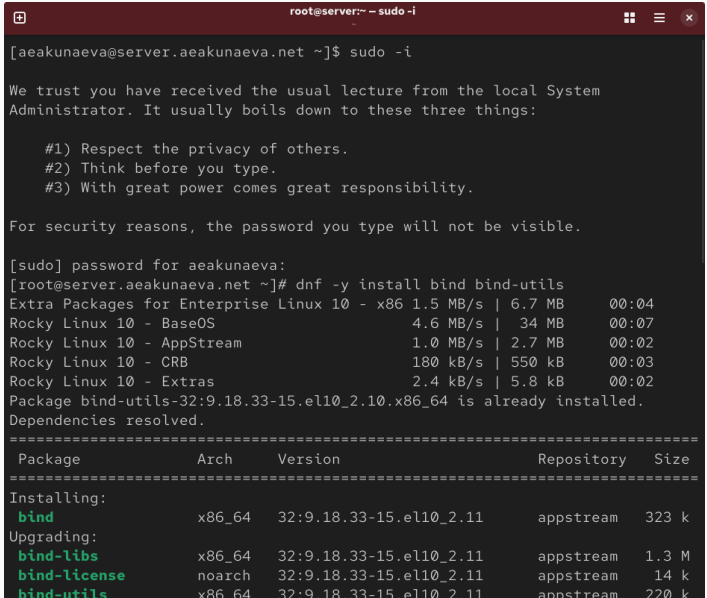
- Linux (дистрибутив Rocky 10.2)
- Linux Fedora Workstation (Markdown)
- Oracle VirtualBox 7.2.16
- Vagrant 2.4.9 (Windows amd64)
- FAR Manager 3.0 build 6699 x64
- Packer 1.16.0 (Windows amd64)

Выполнение лабораторной работы

Запуск виртуальной машины server



Установка bind и bind-utils



Запрос dig к DNS-адресу

```
[root@server.aeakunaeva.net ~]# dig www.yandex.ru~

; <<>> DiG 9.18.33 <<>> www.yandex.ru~
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 24761
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;www.yandex.ru~.                        IN      A

;; AUTHORITY SECTION:
.                       86400   IN      SOA     a.root-servers.net. nstld.verisign-grs.com. 2026
100300 1800 900 604800 86400

;; Query time: 224 msec
;; SERVER: 10.0.0.2#53(10.0.0.2) (UDP)
;; WHEN: Sat Oct 03 15:36:02 UTC 2026
;; MSG SIZE  rcvd: 118
```

Установка DNS-сервера сервером по умолчанию

```
[root@server.aeakunaeva.net ~]# nmcli connection edit eth0

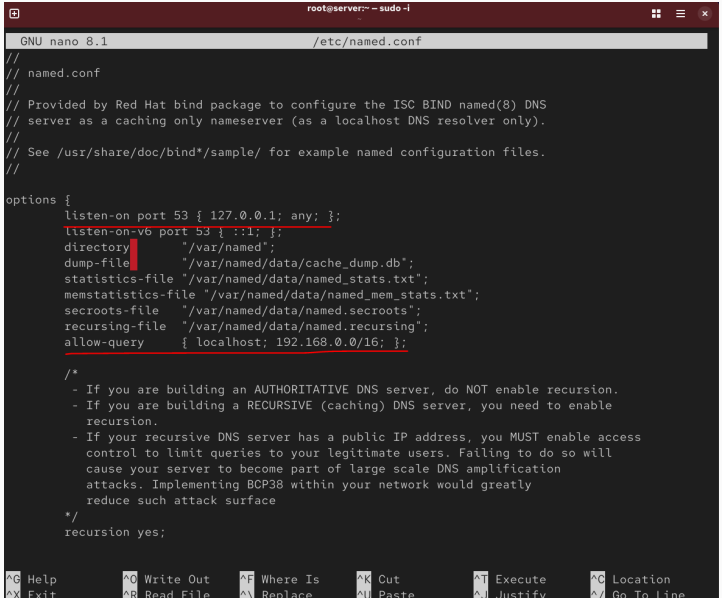
===| nmcli interactive connection editor |===

Editing existing '802-3-ethernet' connection: 'eth0'

Type 'help' or '?' for available commands.
Type 'print' to show all the connection properties.
Type 'describe [<setting>.<prop>]' for detailed property description.

You may edit the following settings: connection, 802-3-ethernet (ethernet), 802-1x, dcb, sriov,
ethtool, match, ipv4, ipv6, prefix-delegation, hostname, link, tc, proxy
nmcli> remove ipv4.dns
nmcli> set ipv4.ignore-auto-dns yes
nmcli> set ipv4.dns 127.0.0.1
nmcli> save
Connection 'eth0' (5a4401f2-dbc3-4baa-b1aa-d9d8725cc78a) successfully updated.
nmcli> quit
[root@server.aeakunaeva.net ~]# nmcli connection edit System\ eth0
Error: Unknown connection 'System eth0'.
[root@server.aeakunaeva.net ~]# systemctl restart NetworkManager
[root@server.aeakunaeva.net ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search aeakunaeva.net
nameserver 127.0.0.1
nameserver fd17:625c:f037:2::3
[root@server.aeakunaeva.net ~]# nano /etc/named.conf
[root@server.aeakunaeva.net ~]#
```

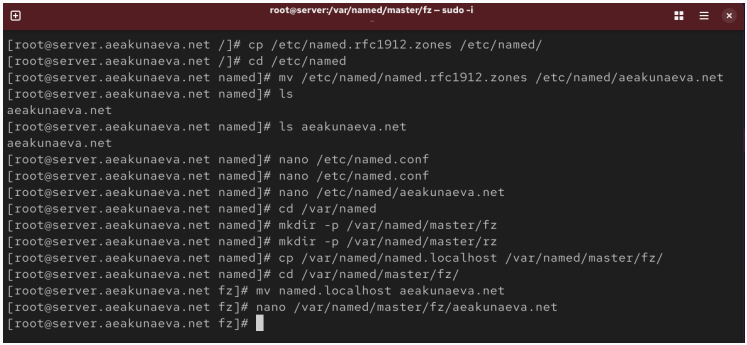
Настройка направления DNS-запросов



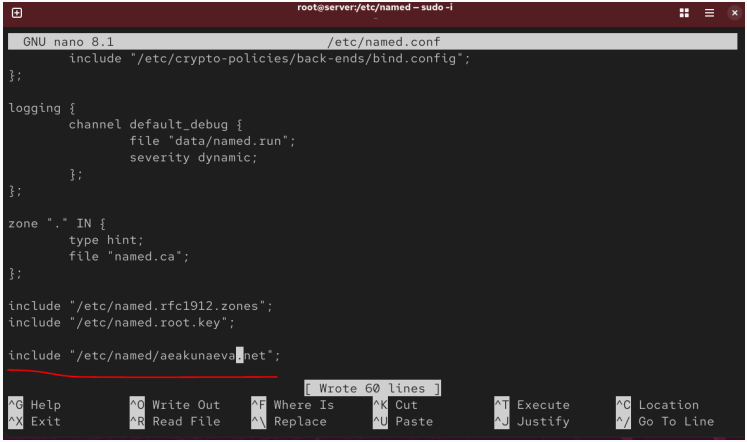
Разрешение работы с DNS

```
[root@server.aeakunaeva.net ~]# firewall-cmd --add-service=dns
success
[root@server.aeakunaeva.net ~]# firewall-cmd --add-service=dns --permanent
success
[root@server.aeakunaeva.net ~]# lsof | grep UDP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1001/gvfs
      Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1001/doc
      Output information may be incomplete.
avahi-dae  1064                          avahi   12u     IPv4              10460       0t0
  UDP *:mdns
avahi-dae  1064                          avahi   13u     IPv6              10461       0t0
  UDP *:mdns
chronyd    1065                         chrony    4u     IPv4               9660       0t0
  UDP localhost:323
chronyd    1065                         chrony    5u     IPv6               9661       0t0
  UDP localhost:323
named      14088                          named   25u     IPv4              93568       0t0
  UDP localhost:domain
named      14088                          named   26u     IPv4              93569       0t0
  UDP localhost:domain
named      14088                          named   31u     IPv6              93572       0t0
  UDP localhost:domain
named      14088                          named   32u     IPv6              93573       0t0
  UDP localhost:domain
named      14088 14089 isc-net-0          named   25u     IPv4              93568       0t0
  UDP localhost:domain
named      14088 14089 isc-net-0          named   26u     IPv4              93569       0t0
  UDP localhost:domain
named      14088 14089 isc-net-0          named   31u     IPv6              93572       0t0
  UDP localhost:domain
named      14088 14089 isc-net-0          named   32u     IPv6              93573       0t0
  UDP localhost:domain
named      14088 14090 isc-net-0          named   25u     IPv4              93568       0t0
  UDP localhost:domain
named      14088 14090 isc-net-0          named   26u     IPv4              93569       0t0
  UDP localhost:domain
named      14088 14090 isc-net-0          named   31u     IPv6              93572       0t0
  UDP localhost:domain
named      14088 14090 isc-net-0          named   32u     IPv6              93573       0t0
  UDP localhost:domain
named      14088 14091 isc-net-0          named   25u     IPv4              93568       0t0
  UDP localhost:domain
named      14088 14091 isc-net-0          named   26u     IPv4              93569       0t0
  UDP localhost:domain
```

Проверка потока DNS-запросов через узел server



Создание шаблонов описания DNS-зон



Включение файла описания зоны /etc/named/aeakunaeva.net

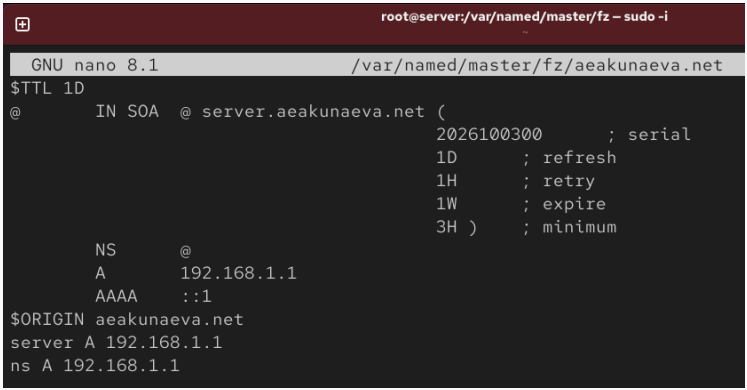
```
  GNU nano 8.1                          /etc/named/aeakunaeva.net
// named.rfc1912.zones:
//
// Provided by Red Hat caching-nameserver package
//
// ISC BIND named zone configuration for zones recommended by
// RFC 1912 section 4.1 : localhost TLDs and address zones
// and https://tools.ietf.org/html/rfc6303
// (c)2007 R W Franks
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//
// Note: empty-zones-enable yes; option is default.
// If private ranges should be forwarded, add
// disable-empty-zone "."; into options
//

zone "aeakunaeva.net" IN {
        type master;
        file "master/fz/aeakunaeva.net";
        allow-update { none; };
};

zone "1.168.192.in-addr.arpa" IN {
        type master;
        file "master/rz/192.168.1";
        allow-update { none; };
};

                                [ Wrote 27 lines ]
^G Help       ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Lo
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go
```

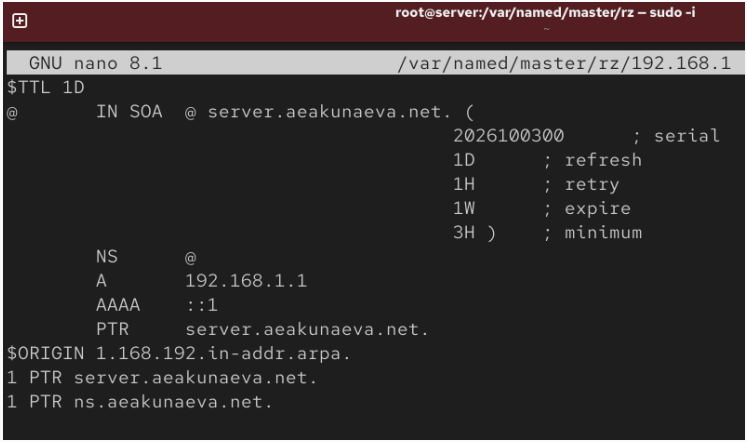
Замена зон в /etc/named/aeakunaeva.net



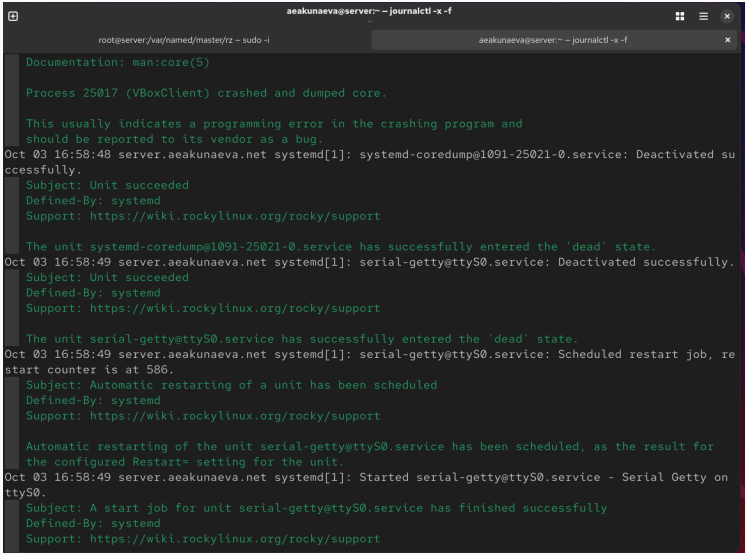
Редактирование данных прямой зоны

```
[root@server.aeakunaeva.net fz]# cp /var/named/named.loopback /var/named/master/rz/
[root@server.aeakunaeva.net fz]# cd /var/named/master/rz/
[root@server.aeakunaeva.net rz]# mv named.loopback 192.168.1
[root@server.aeakunaeva.net rz]# nano /var/named/master/rz/192.168.1
[root@server.aeakunaeva.net rz]# chown -R named:named /etc/named
[root@server.aeakunaeva.net rz]# chown -R named:named /var/named
[root@server.aeakunaeva.net rz]# restorecon -vR /etc
Relabeled /etc/lvm/devices/system.devices from system_u:object_r:lvm_metadata_t:s0 to system_u:o
bject_r:lvm_etc_t:s0
Relabeled /etc/lvm/devices/backup/system.devices-20260912.165102.0005 from system_u:object_r:lvm
_metadata_t:s0 to system_u:object_r:lvm_etc_t:s0
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:us
er_tmp_t:s0 to unconfined_u:object_r:NetworkManager_etc_rw_t:s0
Relabeled /etc/foomatic/hashes.d/hashes.new from system_u:object_r:etc_runtime_t:s0 to system_u:
object_r:etc_t:s0
[root@server.aeakunaeva.net rz]# restorecon -vR /var/named
[root@server.aeakunaeva.net rz]# getsebool -a | grep named
named_tcp_bind_http_port --> off
named_write_master_zones --> on
[root@server.aeakunaeva.net rz]# setsebool named_write_master_zones 1
[root@server.aeakunaeva.net rz]# setsebool -P named_write_master_zones 1
[root@server.aeakunaeva.net rz]# systemctl restart named
```

Настройка меток DNS-зон и конфигурационных файлов



Редактирование данных обратной зоны



Расширенный лог системных сообщений

```
[root@server.aeakunaeva.net rz]# dig ns.aeakunaeva.net

; <<>> DiG 9.18.33 <<>> ns.aeakunaeva.net
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 46222
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: d873569cce7838c9010000006ac13501ccb9825c5e64cf51 (good)
;; QUESTION SECTION:
;ns.aeakunaeva.net.             IN      A

;; AUTHORITY SECTION:
aeakunaeva.net.         10800   IN      SOA     aeakunaeva.net. server.aeakunaeva.net.aeakunaeva.net. 0
 86400 3600 604800 10800

;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Sat Oct 03 17:01:53 UTC 2026
;; MSG SIZE  rcvd: 146

[root@server.aeakunaeva.net rz]# host -l aeakunaeva.net
aeakunaeva.net name server aeakunaeva.net.
aeakunaeva.net has address 192.168.1.1
aeakunaeva.net has IPv6 address ::1
ns.aeakunaeva.net.aeakunaeva.net has address 192.168.1.1
server.aeakunaeva.net.aeakunaeva.net has address 192.168.1.1
[root@server.aeakunaeva.net rz]# host -a aeakunaeva.net
Trying "aeakunaeva.net"
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 31337
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;aeakunaeva.net.                        IN      ANY

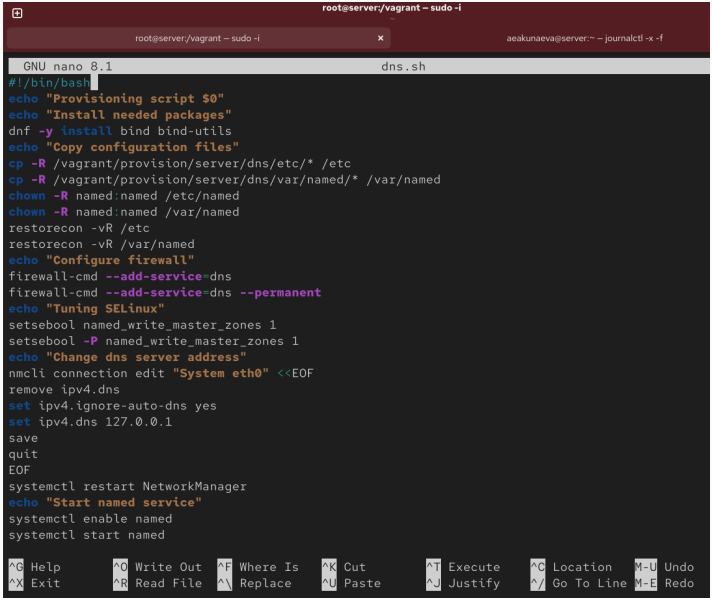
;; ANSWER SECTION:
aeakunaeva.net.         86400   IN      SOA     aeakunaeva.net. server.aeakunaeva.net.aeakunaeva.net. 0
 86400 3600 604800 10800
aeakunaeva.net.         86400   IN      NS      aeakunaeva.net.
aeakunaeva.net.         86400   IN      A       192.168.1.1
aeakunaeva.net.         86400   IN      AAAA    ::1
```

Внесение изменений в еастройки внутреннего окружения server

```
[root@server.aeakunaeva.net rz]# cd /vagrant
[root@server.aeakunaeva.net vagrant]# mkdir -p /vagrant/provision/server/dns/etc/named
[root@server.aeakunaeva.net vagrant]# mkdir -p /vagrant/provision/server/dns/var/named/master/
[root@server.aeakunaeva.net vagrant]# cp -R /etc/named.conf /vagrant/provision/server/dns/etc/
[root@server.aeakunaeva.net vagrant]# cp -R /etc/named/* /vagrant/provision/server/dns/etc/named/

[root@server.aeakunaeva.net vagrant]#
[root@server.aeakunaeva.net vagrant]# cp -R /var/named/master/*
/vagrant/provision/server/dns/var/named/master/
-bash: /vagrant/provision/server/dns/var/named/master/: Is a directory
[root@server.aeakunaeva.net vagrant]# cp -R /var/named/master/* /vagrant/provision/server/dns/var/named
/master/
[root@server.aeakunaeva.net vagrant]# touch dns.sh
[root@server.aeakunaeva.net vagrant]# chmod +x dns.sh
[root@server.aeakunaeva.net vagrant]# nano dns.sh
[root@server.aeakunaeva.net vagrant]#
```

Скрипт dns.sh для установки и настройки DNS-сервера



Внесение изменений в Vagrantfile

Vagrantfile

C: > work > aeakunaeva > vagrant > Vagrantfile

```
4   Vagrant.configure("2") do |config|
57    config.vm.define "server", autostart: false do |server|
67      server.vm.network :private_network,
70
71      server.vm.provider :virtualbox do |virtualbox|
72        virtualbox.customize ["modifyvm", :id, "--vrde", "on"]
73        virtualbox.customize ["modifyvm", :id, "--vrdeport", "3391"]
74      end
75
76      server.vm.provision "server dummy",
77                          type: "shell",
78                          preserve_order: true,
79                          path: "provision/server/01-dummy.sh"
80
81      server.vm.provision "server dns",
82                          type: "shell",
83                          preserve_order: true,
84                          path: "provision/server/dns.sh"
85    end
86
```

Выводы

Выводы

Я приобрела практические навыки по установке и конфигурированию DNS-сервера, усвоила принципы работы системы доменных имён.